

Distributed Cooperative Defense Against DDoS Attacks in Edge-Cloud Computing Networks: A Game-Theoretic Approach

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Abstract—As a promising computing paradigm, edge-cloud computing network integrates the ubiquitous computing resources on the cloud and edge servers to provide high-quality services, but is susceptible to complicated distributed denial-of-service (DDoS) attacks. Although some works have studied edge DDoS mitigation, the cooperative defense among cloud and edge servers considering the DDoS attacker's strategic attack strategy is yet to be explored. In this paper, we model the interactions between the DDoS attacker and defenders as a two-stage dynamic game and propose a distributed cooperative defense scheme. In Stage I, the DDoS attacker strategically launches different amounts of malicious traffic to different edge servers to maximize the total filtering cost. In Stage II, edge servers under attack (i.e., defenders) filter their traffic on the cloud and other edge servers cooperatively to minimize the total filtering cost. The defenders' problem in Stage II is NP-hard and we solve the problem by modeling defenders' behaviors as a selfish filtering game. We prove that the selfish filtering game admits a unique Nash equilibrium (NE) with guaranteed social efficiency, and design both a centralized algorithm and a distributed algorithm to calculate the NE. For the attacker's problem in Stage I, we first analyze a special case where each edge server has the same amount of normal traffic, and design a low-complexity algorithm to calculate the optimal attack strategy. We then analyze the general case where edge servers have different amounts of normal traffic, for which we derive the approximate optimal solution. Simulation results show that the DDoS attacker tends to launch attacks to all edge servers to reduce their cooperative defense capability, and our proposed distributed cooperative defense

mechanism can effectively reduce the total filtering cost compared with existing benchmark defense mechanisms.

Index Terms—Edge-cloud computing networks, DDoS attacks, cooperative defense, game analysis.

I. INTRODUCTION

THE edge-cloud computing network [1], [2], [3] emerges as a novel promising computing paradigm that provides computing services with ultra-low latency and high computing capability. By integrating and allocating the ubiquitous computing resources from cloud and edge servers properly, the edge-cloud computing network can provide a high quality of experience for various applications like autonomous driving, smart healthcare, smart factory, and virtual reality. Due to its great market and application value, the edge-cloud computing network has attracted significant attention from researchers and industry. For example, Google proposes Google Distributed Cloud Edge, Microsoft launches Azure Edge Zones, and Huawei provides IoTEdge.

The distinguishing features, e.g., the geographical distribution and the heterogeneous computing resources of edge servers, make the edge-cloud computing network vulnerable to the distributed denial-of-service (DDoS) attacks [4]. Specifically, the DDoS attacker sends malicious traffic to the network to exhaust edge servers' limited computing resources and prevent the edge-cloud computing network from offering high-quality computing services [5], [6], [7]. To defend against DDoS attacks, edge servers under attack, which we call defenders, need to filter the suspicious (including both normal and malicious) traffic to eliminate the malicious traffic. This reduces the available computing resources to execute users' tasks and causes significant economic losses. In 2022, Huawei [8] reported 104,922 attacks exceeding 100 Gbps in networks of cloud and heterogeneous edge servers, which is a 1.5-fold increase compared with 2021. Furthermore, Zayo reported that organizations without adequate protection suffer from an average cost of \$200,000 per DDoS attack [9].

To mitigate DDoS attacks, researchers have proposed defense approaches including DDoS mitigation hardware, cloud scrubbing services, and edge DDoS mitigation. The hardware-based DDoS mitigation approach installs the firewall and intrusion prevention system, which is costly to deploy and upgrade. Furthermore, application-level DDoS attacks can evade firewall detection [10]. The cloud scrubbing service, offered by large cloud providers such as Cloudflare and Ali Cloud, filters victim's suspicious traffic to eliminate

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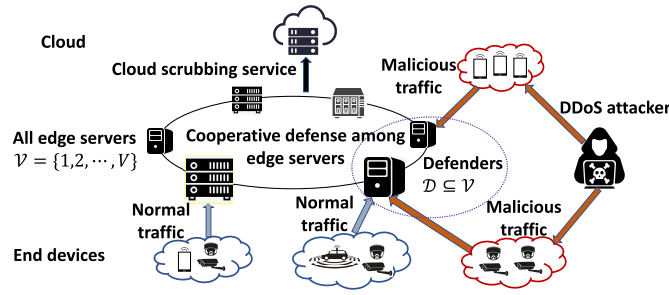


Fig. 1. DDoS attack and defense in the edge-cloud computing network.

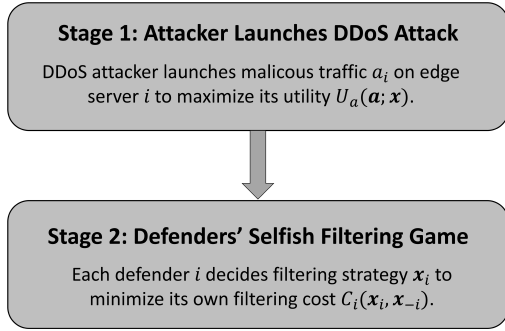


Fig. 2. Two-stage dynamic game.

malicious traffic and redirect normal traffic back to the victim [11]. However, detouring all victims' traffic to cloud for scrubbing may incur high transmission latency. In recent years, researchers have proposed edge DDoS mitigation [12] where edge servers launch virtual DDoS defense instances based on network function virtualization (NFV) [13] and software defined network (SDN) [14]. However, edge DDoS mitigation may suffer from performance degradation when the attacker launches massive DDoS attacks to all edge servers simultaneously [15]. Consequently, there is a critical need to design effective DDoS defense mechanisms to deal with the complicated DDoS attacks in edge-cloud computing networks.

Although edge-cloud computing networks have attracted wide attention, few research studies the interactions between the DDoS attacker and defenders in edge-cloud computing networks, which is the premise for designing effective DDoS defense mechanisms. As shown in Fig. 1, the edge-cloud computing network connects the cloud and heterogeneous edge servers, and the DDoS attacker can compromise some end devices and control them to send malicious traffic to edge servers. We model the interactions between the DDoS attacker and edge servers under attack (i.e., defenders) as a two-stage dynamic game as shown in Fig. 2. In Stage I, the DDoS attacker strategically launches different amounts of malicious traffic to different edge servers in the network. In Stage II, defenders decide their traffic filtering strategies, i.e., how to filter their traffic using the traffic scrubbing/filtering services on the cloud and edge servers. Analyzing the two-stage dynamic game is challenging since the edge-cloud computing network potentially involves a large number of heterogeneous edge servers, which makes the DDoS attack strategy quite

complicated. Furthermore, the strategies of the DDoS attacker and defenders are coupled in a non-trivial way. This motivates us to answer the first key question in edge-cloud computing networks:

Key Question 1: How do the DDoS attacker and defenders strategically interact in the edge-cloud computing network?

To cope with the DDoS attacker's strategic attack, we propose a cooperative defense scheme for the edge-cloud computing network, which fully utilizes the heterogeneous traffic scrubbing/filtering capabilities of the connected cloud and edge servers. Specifically, edge servers under attack, i.e., defenders, can transfer part of their traffic to other edge servers and the cloud to help them filter traffic. Note that each edge server has limited computing resources, and filtering a large amount of traffic on a single edge server may cause congestion which leads to filtering latency. Additionally, transferring massive traffic from edge servers to the cloud for scrubbing may incur transmission latency. Since the latency of traffic filtering directly affects users' experiences, we model the filtering cost as a function of the filtering latency [15], [16]. The goal of the cooperative defense scheme is to optimize defenders' filtering strategies to minimize the total filtering cost in the network, which is NP-hard. We solve this problem in a distributed manner by modeling defenders' filtering behaviors as a selfish filtering game where each defender decides its filtering strategy to minimize its own filtering cost. Analyzing the equilibrium of the game is challenging considering the filtering congestion on heterogeneous edge servers. This leads to the second key question in edge-cloud computing networks:

Key Question 2: How can the edge-cloud computing network defend against DDoS attacks effectively by fully utilizing the heterogeneous traffic scrubbing/filtering capabilities in the network?

In this paper, we study the distributed cooperative defense scheme to mitigate the strategic DDoS attacks in potentially large-scale edge-cloud computing networks effectively and efficiently. We summarize the main contributions of our work as follows.

- **Novel Distributed Cooperative Defense Framework Against Strategic DDoS Attacks:** To the best of our knowledge, we propose the first distributed cooperative defense scheme against strategic DDoS attacks in potentially large-scale edge-cloud computing networks. It aims to mitigate DDoS attacks effectively and efficiently by fully utilizing the heterogeneous traffic scrubbing/filtering capabilities of the cloud and edge servers in the network.
- **Defenders' Selfish Filtering Game:** Minimizing the total filtering cost in the network is NP-hard, and we solve it by modeling a selfish filtering game where each defender minimizes its own filtering cost. Analyzing the equilibrium is challenging due to edge servers' heterogeneity and congestion. We prove that the game admits a unique Nash equilibrium, under which defenders fall into three types and the filtering latency on each edge server is the same. We design a distributed algorithm to calculate the equilibrium filtering strategy without knowing the global

information of the network, and prove that the derived NE has guaranteed social efficiency.

- **DDoS Attacker's Strategic Attack Optimization:** For the attacker's optimization problem, we first analyze a special case where edge servers have the same normal traffic. We show that launching the same amount of malicious traffic to edge servers of the same type at equilibrium is an optimal attack strategy, and we design a low-complexity algorithm to derive it. For the general case where edge servers have different normal traffic, we show that the attack strategy under which edge servers' total traffic ranking remains the same as the normal traffic ranking is an optimal solution, and we design an algorithm to derive the approximate optimal solution.
- **Simulation Results and Practical Insights:** Simulation results show that the DDoS attacker tends to launch attacks to all edge servers to reduce their cooperative defense capability, and our proposed distributed cooperative defense mechanism can effectively reduce the total filtering cost compared with existing benchmark defense mechanisms. Our distributed cooperative defense scheme captures the strategic attacks of the attacker, the heterogeneity of edge servers, and the cooperation among the cloud and edge servers, which makes our derived insights and results more practically significant.

We organize the rest of this paper as follows. We review related literature in Section II and introduce the system model in Section III. We analyze defenders' traffic filtering behaviors in Section IV and analyze the attack behavior of the attacker in Section V. We provide simulation results in Section VI and conclude our work in Section VII.

II. RELATED WORK

There have been extensive studies on defense mechanisms against attacks in networks. In this section, we discuss the related works regarding game analysis of attack-defense interactions and cooperative defense against DDoS attacks.

A. Game Analysis of Attack-Defense Interactions

Game theory has been widely used to analyze the interactions between the attacker and the defender in the field of cyber security. Chen et al. [7] develop a Stackelberg game-based collaborative framework for IoT DDoS defense, where the attacker allocates resources across multiple control points to attack only a single victim. Wahab et al. [17] and Li et al. [18] model Bayesian Stackelberg games between the Advance Persistence Threat (APT) attacker and the cloud/edge defender to deal with the unknown types of attacks, where the attacker only attacks one target or attacks all targets at the same time. Xiao et al. [19] use prospect theory to model the interactions between an APT attacker and a cloud defender with bounded rationality. Zhou et al. [20] propose a defender-led signaling game to model the interactions between the defender of the IoT network and the DDoS attacker who can only choose to attack or not attack. Liu et al. [21] propose a Bayesian Q-learning game to model the interactions between a cooperative sensor-edgeVM pair and a DDoS attacker under

incomplete information, where the attacker targets only a single defender.

We can see that most existing works greatly simplify the attacker's action. For example, the attacker can only decide whether to attack, or the attacker can only attack one target, or the attacker can only attack all targets with the same strategy. In practice, however, launching different degrees of attacks to different targets strategically can compromise the system more effectively. In this paper, we model such strategic attack behavior of the DDoS attacker in the edge-cloud computing network with heterogeneous edge servers.

B. Cooperative Defense Against DDoS Attacks

There have been some works studying cooperative defense against DDoS attacks. You et al. [11] investigate the scheduling of DDoS flows through an online auction mechanism, utilizing the scrubbing centers. Li et al. [12] design a cooperative defense framework which coordinates container-carried defense resources among edge servers through online combinatorial auction to mitigate the application-level DDoS attacks. Myneni et al. [22] propose a distributed defense solution against DDoS attacks, which is deployed at both customer and provider edges, employing neural network techniques. Feng et al. [23] propose a reinforcement learning-based collaborative DDoS detection method and design a lightweight unsupervised classifier based on statistics at the edge of IoT. He et al. [16] propose a collaborative edge DDoS mitigation scheme where edge servers can transfer their requests to adjacent edge servers to mitigate DDoS attacks. Deng et al. [24] develop a graph neural network (GNN)-based collaborative deep reinforcement learning model to generate the resource provisioning and mitigating strategy for DDoS attacks in mobile edge computing-enabled vehicular ad hoc networks. Huang et al. [25] propose a risk-aware task-offloading framework for edge DDoS mitigation which first evaluates the trust level of edge nodes, then clusters them through a weighted K-means algorithm, and dispatches user tasks to the safest and most capable nodes. Xu et al. [26] focus on bandwidth-layer DDoS defense in mobile edge computing and propose a evolutionary-game model for user-bandwidth allocation. Zhou et al. [27] study stealthy low-rate DDoS defense and propose a deep reinforcement learning-based approach.

However, previous studies usually focus on edge and cloud DDoS defenses independently, rather than exploring their integration. They do not consider the congestion effect of edge filtering and ignore the traffic scrubbing service on the cloud. Note that filtering massive malicious traffic on one edge server may result in congestion and incur high filtering latency, potentially leading to denial of service. In this case, redirecting a portion of suspicious traffic from congested edge servers to a cloud scrubbing center may alleviate the latency issue and effectively mitigate DDoS attacks.

In this paper, we propose a distributed cooperative defense scheme, which fully utilizes the heterogeneous traffic scrubbing/filtering capabilities of cloud and edge servers, to mitigate the strategic DDoS attacks in potentially large-scale edge-cloud computing networks. Our analysis can guide the

edge-cloud computing network to tackle the DDoS attack problem effectively and efficiently.

III. SYSTEM MODEL

In this section, we introduce the system model. As shown in Fig. 1, we consider an edge-cloud computing network which integrates the computing resources of a cloud and a set $\mathcal{V} = \{1, 2, \dots, V\}$ of heterogeneous edge servers and provides computing services with ultra-low latency to end devices. End devices send computing requests (i.e., normal traffic) such as autonomous driving tasks to these edge servers. However, the edge-cloud computing network faces DDoS attack threats. We consider application-level DDoS attacks [12], where the attacker can compromise some end devices and control them to strategically launch malicious application-level DDoS attacks to edge servers, aiming to exhaust edge servers' limited computing resources and prevent the edge-cloud computing network from offering high-quality computing services. To mitigate DDoS attacks effectively, edge servers under attack (i.e., defenders) can filter their suspicious traffic (including both the normal and malicious traffic) through traffic scrubbing on the cloud and traffic filtering on edge servers in the network cooperatively, to eliminate the malicious traffic. We model the interactions of the DDoS attacker and defenders as a two-stage dynamic game as shown in Fig. 2. In the rest of the section, we first model defenders' cooperative defense problem in Stage II, and then model the DDoS attacker's attack problem in Stage I.

A. Defenders' Cooperative Defense Problem

The edge-cloud computing network can design the cooperative traffic filtering strategies to minimize the total filtering cost in the network, i.e., the traffic filtering costs of all edge servers under attack. We define the defender set $\mathcal{D} = \{1, 2, \dots, D\}$ as the set of edge servers that are under DDoS attack. We can see that $\mathcal{D} \subseteq \mathcal{V}$. For a defender $i \in \mathcal{D}$, its suspicious traffic includes the normal traffic λ_i (in bits) sent by benign end devices and the malicious traffic a_i (in bits) sent by compromised end devices. Since application-level DDoS attacks can exhaust a large amount of computing resources by sending only a few requests, defenders should filter all suspicious traffic to mitigate DDoS attacks effectively. Under our proposed cooperative defense scheme for the edge-cloud computing network, defender i can filter a proportion x_{ik} of its suspicious traffic on each edge server $k \in \mathcal{V}$ and a proportion x_{ic} of its suspicious traffic on the cloud. We denote $\mathbf{x}_i = \{x_{ic}, x_{ik}, \forall k \in \mathcal{V}\}$ as the filtering strategy of defender i , and denote $\mathbf{x} = \{\mathbf{x}_i, \forall i \in \mathcal{D}\}$ as all defenders' filtering strategies. We further denote $\mathbf{a} = \{a_i, \forall i \in \mathcal{V}\}$ as the DDoS attacker's attack strategy. In the following, we first model each defender's filtering cost, which includes the filtering cost on edge servers and the filtering cost on the cloud. We then model the total filtering cost minimization problem of the edge-cloud computing network.

1) *Defender's Filtering Cost:* For each defender, filtering its suspicious traffic on edge servers and on the cloud both incur filtering costs.

a) *Filtering Cost on Edge Servers:* Since the latency of traffic filtering directly affects end devices' experiences, we model the filtering cost as a function of the filtering latency. When defender i filters part of its traffic on edge server $k \in \mathcal{V}$, the filtering latency is mainly due to edge server k 's limited computing resources for traffic filtering and the congestion effect caused by resource sharing among defenders who filter traffic on edge server k .¹ We denote the traffic filtering capability of edge server k as f_k (in bits/second), i.e., the amount of traffic that edge server k can filter per second [2], [28].² In this case, the traffic filtering latency on edge server k is

$$F_k(\mathbf{x}; \mathbf{a}) = \frac{\sum_{i \in \mathcal{D}} (\lambda_i + a_i) x_{ik}}{f_k}, \quad (1)$$

where $\sum_{i \in \mathcal{D}} (\lambda_i + a_i) x_{ik}$ is the total amount of suspicious traffic that defenders filter on edge server k . We model the filtering cost on edge servers as the weighted filtering latency on chosen edge servers [30], which is

$$C_i^e(\mathbf{x}; \mathbf{a}) = \sum_{k \in \mathcal{V}} x_{ik} F_k(\mathbf{x}; \mathbf{a}). \quad (2)$$

b) *Filtering Cost on the Cloud:* When a defender transfers part of its traffic to the cloud for traffic scrubbing, the latency mainly comes from transmission delay. Note that since the cloud scrubbing center usually has sufficient computing resources, the latency of traffic scrubbing is negligible.³ For defender i , we model its filtering cost on the cloud as the transmission latency, which is

$$C_i^c(x_{ic}; a_i) = w_c(\lambda_i + a_i)x_{ic}, \quad (3)$$

where w_c (in seconds/bit) is a coefficient representing the transmission latency for transferring a unit of traffic to the cloud [32], [33].

We define the filtering cost of defender i as the sum of the filtering costs on edge servers and cloud, which is

$$C_i(\mathbf{x}_i, \mathbf{x}_{-i}; \mathbf{a}) = C_i^e(\mathbf{x}; \mathbf{a}) + C_i^c(x_{ic}; a_i). \quad (4)$$

Here $\mathbf{x}_{-i} = \{\mathbf{x}_j, \forall j \in \mathcal{D} \setminus \{i\}\}$ represents the filtering strategies of all other defenders except defender i .

2) *Total Filtering Cost Minimization Problem:* To mitigate the DDoS attacks effectively, the goal of the edge-cloud computing network is to minimize the total filtering cost in the network, i.e., the traffic filtering costs of all edge servers under attack. We model the total filtering cost minimization problem as follows,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \sum_{i \in \mathcal{D}} C_i(\mathbf{x}_i; \mathbf{a}) \\ \text{s.t.} \quad & \sum_{k \in \mathcal{V}} x_{ik} + x_{ic} = 1, \forall i \in \mathcal{D}, \\ & x_{ik}, x_{ic} \geq 0, \forall k \in \mathcal{V}, i \in \mathcal{D}. \end{aligned} \quad (5)$$

¹We consider edge servers connected through high-speed links, and the latency of transferring traffic among edge servers is negligible compared with the latency of traffic filtering [28].

²For example, Huawei's AntiDDoS1900 measures its traffic filtering capability by the amount of traffic that it can filter per second [29].

³For example, Ali Cloud [31] can filter suspicious traffic of Tb/s level.

The first constraint indicates that each defender should filter all its suspicious traffic. The second constraint restricts that the proportion of traffic to filter on an edge server or the cloud should be non-negative. We later solve the total filtering cost minimization problem in Section IV.

B. DDoS Attacker's Attack Problem

The DDoS attacker strategically launches different amounts of malicious traffic to different edge servers, aiming to exhaust the limited computing resources of heterogeneous edge servers and prevent the edge-cloud computing network from offering high-quality computing services. Recall that we denote a_i as the amount of malicious traffic that the DDoS attacker launches to edge server $i \in \mathcal{V}$, and denote $\mathbf{a} = \{a_i, \forall i \in \mathcal{V}\}$ as the DDoS attacker's attack strategy. Since the DDoS attacker usually has limited resource for attack, we assume that the total amount of malicious traffic that the attacker can launch to the edge-cloud computing network has an upper bound a_{max} , i.e., $\sum_{i \in \mathcal{V}} a_i \leq a_{max}$. We model the objective of the DDoS attacker as maximizing the total filtering cost of the edge-cloud computing network, i.e.,

$$U_a(\mathbf{a}; \mathbf{x}) = \sum_{i \in \mathcal{D}} C_i(\mathbf{x}; \mathbf{a}). \quad (6)$$

Then we formulate the optimization problem of the DDoS attacker as follows,

$$\begin{aligned} \max_{\mathbf{a}} \quad & U_a(\mathbf{a}; \mathbf{x}) \\ \text{s.t.} \quad & \sum_{i \in \mathcal{V}} a_i \leq a_{max}, \\ & a_i \geq 0, \forall i \in \mathcal{V}. \end{aligned} \quad (7)$$

The first constraint indicates that the total amount of malicious traffic launched to the edge-cloud computing network must be within the budget. The second constraint restricts that the amount of malicious traffic launched to each edge server must be non-negative. We further denote the feasible region of the attack strategy as $\mathcal{A} = \{(a_1, a_2, \dots, a_V) | \sum_{i \in \mathcal{V}} a_i \leq a_{max}, a_i \geq 0, \forall i \in \mathcal{V}\}$.

We later solve the two-stage dynamic game between the DDoS attacker and defenders by backward induction. We first solve the defenders' cooperative defense problem in Section IV, and then solve the DDoS attacker's attack problem in Section V.

IV. DISTRIBUTED COOPERATIVE DEFENSE MECHANISM

In this section, we aim to design an effective and efficient cooperative defense mechanism for the edge-cloud computing network. We first prove that the total filtering cost minimization problem (5) is NP-hard.

Theorem 1: The total filtering cost minimization problem in the edge-cloud computing network is NP-hard.

Proof: See Appendix A in the supplementary material. $\square$

We later solve the total filtering cost minimization problem in a distributed manner by modeling defenders' traffic filtering behaviors as a selfish filtering game where each defender decides its traffic filtering strategy to minimize its own filtering

cost (in Section IV-A). We analyze the properties of the Nash equilibrium (in Section IV-B) and design both centralized and distributed algorithms to derive the NE (in Section IV-C). We use the NE as the defense strategy of the edge-cloud computing network, and prove that the derived NE has guaranteed performance regarding the total filtering cost (in Section IV-D).

A. Selfish Filtering Game Formulation

We formulate the selfish filtering game among defenders as follows.

Game 1: (Selfish Filtering Game among Defenders)

- *Players:* the set $\mathcal{D}$ of defenders.
- *Strategies:* Each defender $i \in \mathcal{D}$ decides its traffic filtering strategy $\mathbf{x}_i$.
- *Objectives:* Each defender $i \in \mathcal{D}$ aims to minimize its own traffic filtering cost $C_i(\mathbf{x}_i, \mathbf{x}_{-i})$ calculated in (4).

We first define the best response of each defender $i \in \mathcal{D}$ in the selfish filtering game. Given other defenders' strategies $\mathbf{x}_{-i}$, the best response of defender $i \in \mathcal{D}$ is

$$\begin{aligned} BR_i(\mathbf{x}_{-i}) = \arg \min_{\mathbf{x}_i} \quad & C_i(\mathbf{x}_i, \mathbf{x}_{-i}) \\ \text{s.t.} \quad & \sum_{k \in \mathcal{V}} x_{ik} + x_{ic} = 1, \\ & x_{ik}, x_{ic} \geq 0, \forall k \in \mathcal{V}. \end{aligned} \quad (8)$$

The Nash equilibrium of the selfish filtering game is a strategy profile under which each defender plays its best response. We next formally define the Nash equilibrium of Game 1.

Definition 1: (Nash Equilibrium of Game 1) A strategy profile $\mathbf{x}^* = \{\mathbf{x}_i^*, \forall i \in \mathcal{D}\}$ is a Nash equilibrium of the selfish filtering game if each defender $i \in \mathcal{D}$ plays its best response under $\mathbf{x}^*$, i.e., $\mathbf{x}_i^* = BR_i(\mathbf{x}_{-i}^*), \forall i \in \mathcal{D}$.

Under the Nash equilibrium of the selfish filtering game, no defender will change its defense strategy unilaterally. Calculating the NE through solving the best response of each defender is challenging since the best responses of defenders are coupled with each other in a non-trivial way and do not have closed-form solutions. The heterogeneity and congestion effect of edge servers further complicate the equilibrium analysis. We next analyze the properties of the NE which provide insights of defenders' equilibrium behaviors and guide us to calculate the NE.

B. Nash Equilibrium of the Game

Since we use the NE of the selfish filtering game as the defense strategy for the edge-cloud computing network, we first prove the existence of the NE before analyzing the properties of the NE.

Lemma 1: The Nash equilibrium of the selfish filtering game always exists.

Proof: See Appendix B in the supplementary material. $\square$

The main idea is to prove that the selfish filtering game is a concave game, and hence the NE exists [34]. We next analyze some properties of the NE, based on which we prove the uniqueness of the NE.

1) *Properties of the NE*: We characterize the properties of the NE of the selfish filtering game, including the necessary and sufficient condition of the NE, the traffic filtering latency on each edge server at equilibrium, and the relationship between the proportion of traffic that a defender filters on an edge server and the traffic filtering capability of the edge server.

We first characterize the necessary and sufficient condition of the Nash equilibrium by analyzing the KKT conditions of each defender's optimization problem (8).

Lemma 2: Under the Nash equilibrium $\mathbf{x}^*$ of the selfish filtering game, the equilibrium filtering strategy $x_{iu}^* \in \mathbf{x}_i^*$ of each defender $i \in \mathcal{D}$ satisfies

$$\frac{\partial C_i(\mathbf{x}^*)}{\partial x_{iu}^*} \begin{cases} = \min_{x_{ij}^* \in \mathbf{x}_i^*} \frac{\partial C_i(\mathbf{x}^*)}{\partial x_{ij}^*}, & \text{if } x_{iu}^* > 0, \\ \geq \min_{x_{ij}^* \in \mathbf{x}_i^*} \frac{\partial C_i(\mathbf{x}^*)}{\partial x_{ij}^*}, & \text{if } x_{iu}^* = 0. \end{cases}$$

Proof: See Appendix C in the supplementary material. $\square$

Lemma 2 indicates that at equilibrium, defender i filters a proportion of its traffic on an edge server or cloud (i.e., $x_{iu}^* > 0$) only if the marginal filtering cost $\frac{\partial C_i(\mathbf{x}^*)}{\partial x_{iu}^*}$ is the minimum among all other edge servers or cloud. We denote the minimum marginal filtering cost of defender i at equilibrium as β_i , i.e., $\beta_i \triangleq \min_{x_{ij}^* \in \mathbf{x}_i^*} \frac{\partial C_i(\mathbf{x}^*)}{\partial x_{ij}^*}$. When defender i filters a proportion of its traffic on the cloud, we have

$$\beta_i = \frac{\partial C_i(\mathbf{x}^*)}{\partial x_{ic}^*} = w_c(\lambda_i + a_i), \text{ if } x_{ic}^* > 0. \quad (9)$$

When defender i filters a proportion of its traffic on edge server $k \in \mathcal{V}$, we have:

$$\beta_i = \frac{\partial C_i(\mathbf{x}^*)}{\partial x_{ik}^*} = \frac{x_{ik}^*(\lambda_i + a_i)}{f_k} + F_k(\mathbf{x}^*), \text{ if } x_{ik}^* > 0. \quad (10)$$

Based on Lemma 2 and (9) (10), we obtain an important property of the NE regarding the traffic filtering latency $F_k(\mathbf{x}^*)$ on each edge server $k \in \mathcal{V}$.

Theorem 2: Under the NE $\mathbf{x}^*$ of the selfish filtering game, the traffic filtering latency on each edge server is the same, i.e., $F_1(\mathbf{x}^*) = F_2(\mathbf{x}^*) = \dots = F_V(\mathbf{x}^*) = F(\mathbf{x}^*) = F^*$.

Proof: See Appendix D in the supplementary material. $\square$

Intuitively, if an edge server has a lower filtering latency, defenders will switch their traffic from other edge servers with higher filtering latencies to the lower one, leading to uniform filtering latency among edge servers. This indicates that our proposed distributed cooperative defense scheme achieves *load-balanced traffic filtering* among heterogeneous edge servers.

Based on Lemma 2 and Theorem 2, we further obtain a relationship between the proportion of traffic that a defender filters on an edge server and the traffic filtering capability of the edge server in the following lemma.

Lemma 3: Under the NE $\mathbf{x}^*$ of the selfish filtering game, the proportion x_{ik}^* of traffic that defender i filters on each edge server $k \in \mathcal{V}$ is proportional to edge server k 's traffic filtering capability f_k , i.e., $\frac{x_{ik}^*}{f_k} = \frac{x_{ij}^*}{f_j}, \forall k, j \in \mathcal{V}$.

Proof: See Appendix E in the supplementary material. $\square$

Lemma 3 indicates that if a defender filters part of its traffic on an edge server, then it filters its traffic on all edge servers in the network, and the traffic filtered on each edge server is proportional to the edge server's traffic filtering capability.

2) *The Uniqueness of the NE*: To establish the uniqueness of the NE for the selfish filtering game, we first characterize a one-to-one mapping between the filtering latency F^* at equilibrium and the NE $\mathbf{x}^*$. We then prove the uniqueness of the NE by proving the uniqueness of the equilibrium filtering latency F^* .

Recall that given the NE $\mathbf{x}^*$, we can calculate the equilibrium filtering latency $F(\mathbf{x}^*)$ on each edge server according to Eq. (1). In the following theorem, we prove that given the traffic filtering latency F^* at equilibrium, we can calculate the NE $\mathbf{x}^*$. We first sort defenders in set $\mathcal{D}$ by their total amount of normal and malicious traffic (i.e., $\lambda_i + a_i, \forall i \in \mathcal{D}$) in an ascending order. We denote the ranking of defender i as r_i . Then defenders fall into at most three types at equilibrium, depending on two critical rankings $\bar{i}$ and $\bar{j}$.⁴

Theorem 3: Given the equilibrium filtering latency F^* on an edge server, we can calculate the equilibrium filtering strategy $\mathbf{x}^*$, under which defenders fall into one of the following three types.

- *Type I (cloud-only)*: $x_{ic}^* = 1, x_{ik}^* = 0, \forall k \in \mathcal{V}$, if $0 < r_i < \bar{i}$.
- *Type II (mixed-filtering)*: $x_{ik}^* = \frac{w_c(\lambda_i + a_i) - F^*}{\lambda_i + a_i} f_k, \forall k \in \mathcal{V}, x_{ic}^* = 1 - \sum_{k \in \mathcal{V}} x_{ik}^*$, if $\bar{i} \leq r_i < \bar{j}$.
- *Type III (edge-only)*: $x_{ik}^* = \frac{f_k}{\sum_{k \in \mathcal{V}} f_k}, \forall k \in \mathcal{V}, x_{ic}^* = 0$, if $\bar{j} \leq r_i \leq D$.

We calculate the critical ranking $\bar{i}$ as follows.

- If there exist a defender i with ranking r_i and a defender j with ranking $r_j = r_i - 1$ that satisfy $\frac{1}{\lambda_i + a_i} < \frac{w_c}{F^*} \leq \frac{1}{\lambda_j + a_j}$, then $\bar{i} = r_i$.
- Otherwise, $\bar{i} = 1$.

We calculate the critical ranking $\bar{j}$ as follows.

- If the defender s with the smallest traffic satisfies $\frac{1}{\lambda_s + a_s} \leq \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*}$, then $\bar{j} = 1$.
- If the defender m with the largest traffic satisfies $\frac{1}{\lambda_m + a_m} > \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*}$, then $\bar{j} = D + 1$.
- If there exist a defender i with ranking r_i and a defender j with ranking $r_j = r_i - 1$ that satisfy $\frac{1}{\lambda_i + a_i} \leq \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*} < \frac{1}{\lambda_j + a_j}$, then $\bar{j} = r_i$.

Proof: See Appendix F in the supplementary material. $\square$

Theorem 3 shows that at equilibrium, we can classify defenders into at most three types, including the defenders who only filter traffic on the cloud, the defenders who filter traffic on both the cloud and edge servers, and the defenders who only filter traffic on edge servers. Furthermore, if we know the equilibrium filtering latency F^* on each edge server, we can write the closed-form solution of the NE $\mathbf{x}^*$. Note that given the NE $\mathbf{x}^*$, we can also calculate the equilibrium filtering latency $F(\mathbf{x}^*)$ according to Eq. (1). Hence, there

⁴Intuitively, the critical rankings $\bar{i}$ and $\bar{j}$ represent traffic-determined thresholds that categorize defenders into three types based on the comparison between the marginal filtering cost on an edge server and that on the cloud. See Appendix F in the supplementary material for detailed explanations.

is a one-to-one mapping between the equilibrium filtering latency F^* and the NE $\mathbf{x}^*$. We next prove the uniqueness of the NE by proving the uniqueness of F^* .

Theorem 4: The selfish filtering game admits a unique Nash equilibrium.

Proof: See Appendix G in the supplementary material. $\square$

Although the NE is unique, writing a closed-form solution for the NE is challenging since we do not have the closed-form solution for F^* . We next design algorithms to derive the NE $\mathbf{x}^*$ based on the properties of the NE.

C. Algorithms to Calculate the NE

In this subsection, we first design a centralized algorithm with polynomial complexity to derive the NE. We then design a distributed algorithm which does not require the global information of the network to calculate the NE.

1) *A Centralized Algorithm to Calculate the NE:* According to the NE analysis, there is only one pair $(\bar{i}, \bar{j})$ and its corresponding F^* that satisfy the relationships in Theorem 3. Based on this result, we design a centralized algorithm, which first derives the equilibrium filtering latency F^* , and then calculates the NE $\mathbf{x}^*$ according to Theorem 3.

To derive F^* , we explore its relationship with the two critical rankings $\bar{i}$ and $\bar{j}$. Specifically, given F^* , we can calculate $\mathbf{x}^*$ according to Theorem 3, where the derived $\mathbf{x}^*$ should also satisfy the relationship in Eq. (1). In other words, we have

$$F^* = \sum_{r_i=\bar{i}}^{\bar{j}-1} [w_c(\lambda_i + a_i) - F^*] + \sum_{r_i=\bar{j}}^D \frac{\lambda_i + a_i}{\sum_{k \in \mathcal{V}} f_k}.$$

Then we can calculate F^* as a function of the two critical rankings $\bar{i}$ and $\bar{j}$ as follows

$$F^* = \frac{\sum_{r_i=\bar{i}}^{\bar{j}-1} (\lambda_i + a_i) w_c + \sum_{r_i=\bar{j}}^D \frac{1}{\sum_{k \in \mathcal{V}} f_k} (\lambda_i + a_i)}{\bar{j} - \bar{i} + 1}. \quad (11)$$

Recall that the two critical rankings $\bar{i}$ and $\bar{j}$, which are integers in the range from 1 to $D + 1$, also depend on F^* as stated in Theorem 3. Therefore, to derive F^* , we can first enumerate the feasible pair $(\bar{i}, \bar{j})$ under which we calculate F^* according to (11), we then check whether the pair $(\bar{i}, \bar{j})$ and the corresponding F^* satisfy the relationships in Theorem 3. Since F^* is unique, there is only one pair $(\bar{i}, \bar{j})$ and its corresponding F^* that satisfy the relationships in Theorem 3.

We show the centralized algorithm to calculate the NE in Algorithm 1. We first sort defenders by their total amount of normal and malicious traffic, i.e., $\lambda_i + a_i, \forall i \in \mathcal{D}$, in an ascending order to obtain the ranking of each defender (Line 1 in Algorithm 1). We then enumerate the possible pairs of critical rankings $(\bar{i}, \bar{j})$ (Lines 2-3 in Algorithm 1). For each pair, we calculate the equilibrium filtering latency F^* by Eq. (11) (Line 4 in Algorithm 1), and check whether the critical ranking pair $(\bar{i}, \bar{j})$ and the corresponding F^* satisfy the relationship in Theorem 3 (Lines 5-9 in Algorithm 1). If the pair $(\bar{i}, \bar{j})$ and the corresponding F^* satisfy the relationships in Theorem 3, we can use the

Algorithm 1 A Centralized Algorithm to Calculate the NE

```

1: Sort defenders in an ascending order according to  $\lambda_i + a_i, \forall i \in \mathcal{D}$ ;
2: for  $\bar{i} = 1: D$  do
3:   for  $\bar{j} = \bar{i}: D + 1$  do
4:     Calculate  $F^*$  by Eq. (11);
5:     if  $(\bar{i}, \bar{j})$  and  $F^*$  satisfy Theorem 3 then
6:       break;
7:     else
8:       continue;
9:     end if
10:   end for
11: end for
12: Calculate  $\mathbf{x}^*$  by Theorem 3.
```

derived F^* to calculate the NE. If not, we keep on searching for the next possible critical ranking pair.

We prove the convergence and the complexity of Algorithm 1 in the following lemma.

Lemma 4: Algorithm 1 derives the NE of the selfish filtering game with complexity $\mathcal{O}(D^2)$.

Proof: See Appendix H in the supplementary material. $\square$

Since Algorithm 1 can find the NE with low complexity, it can respond to the DDoS attacks efficiently. In practice, however, edge-cloud computing networks can be large networks, and it is impractical for each edge server to obtain the global information, especially when some edge servers are under heavy attack. We next design a distributed algorithm that calculates the equilibrium defense strategy based on each defender's local information.

2) *A Distributed Algorithm to Calculate the NE:* We aim to design a distributed algorithm for each defender to derive its equilibrium filtering strategy without knowing other defenders' filtering strategies and information, which is of significant practical importance in large-scale edge-cloud computing networks. The design of the distributed algorithm depends on the property of the selfish filtering game being a concave game. Specifically, when a concave game admits a unique NE, the best response update process can converge to the NE starting from any feasible strategy profile [34]. Since calculating the NE $\mathbf{x}^*$ is equivalent to calculating the equilibrium filtering latency F^* , the main idea of the distributed algorithm is to let defenders take turns to update the equilibrium filtering latency, considering the effect of their best responses.

We show the distributed algorithm for each defender to calculate its equilibrium filtering strategy in Algorithm 2. Edge servers first announce their filtering capabilities (Line 1 in Algorithm 2). Then defenders send an initial parameter $E_i^0 = \frac{\lambda_i + a_i}{\sum_{k \in \mathcal{V}} f_k}$ to the network operator of the edge-cloud computing network by proportionally filtering all their traffic on all edge servers (Line 2 in Algorithm 2). The network operator sorts defenders by $E_i^0, \forall i \in \mathcal{D}$, and calculates the initial filtering latency F^0 and announces to defenders (Line 3 in Algorithm 2). Then defenders take turns to update the filtering latency. In each iteration t , defender i calculates the increase of its filtering latency after it plays its best

response $x_i^{BR,t}$, which we denote as E_i^t as follows

$$E_i^t = \frac{x_{ik}^{BR,t}(\lambda_i + a_i)}{f_k}. \quad (12)$$

We define $Z_i^t = w_c(\lambda_i + a_i) - (F^t - E_i^{t-1})$. By substituting the best response (calculated by Lemma 2) into (12), defender i can update E_i^t in iteration t as

$$E_i^t = \begin{cases} 0, & \text{if } Z_i^t \leq 0, \\ \frac{Z_i^t}{2}, & \text{if } \frac{Z_i^t}{2(\lambda_i + a_i)} (\sum_{k \in \mathcal{V}} f_k) < 1, \\ \frac{\lambda_i + a_i}{\sum_{k \in \mathcal{V}} f_k}, & \text{if } \frac{Z_i^t}{2(\lambda_i + a_i)} (\sum_{k \in \mathcal{V}} f_k) \geq 1. \end{cases} \quad (13)$$

See Appendix I in the supplementary material for the detailed derivation of E_i^t . After defender i updates its effect E_i^t on the filtering latency, the network operator updates the filtering latency by aggregating all defenders' effects as follows

$$F^t = \sum_{r_j=1}^{r_i} E_j^t + \sum_{r_j=r_i+1}^D E_j^{t-1}, \quad (14)$$

where defenders ranking from 1 to r_i have updated their effects and the others do not. After F^t converges, each defender calculates its equilibrium filtering strategy and filters suspicious traffic (Line 13 in Algorithm 2).

Algorithm 2 A Distributed Algorithm to Calculate the NE

- 1: Each edge server $k \in \mathcal{V}$ announces its filtering capability f_k ;
 - 2: Each defender $i \in \mathcal{D}$ sends $E_i^0 = \frac{\lambda_i + a_i}{\sum_{k \in \mathcal{V}} f_k}$ to the network operator;
 - 3: The network operator sorts defenders by $E_i^0, \forall i \in \mathcal{D}$ in an ascending order, and announces $F^0 = \sum_{i \in \mathcal{D}} E_i^0$ to defenders;
 - 4: Set $t = 0$ and $F^{-1} = -\infty$;
 - 5: **while** $F^t - F^{t-1} \geq \epsilon$ **do**
 - 6: $t = t + 1$;
 - 7: $F^t = F^{t-1}$;
 - 8: **for** defender i with ranking $r_i = 1: D$ **do**
 - 9: Defender i updates E_i^t according to (13) and sends to the network operator;
 - 10: The network operator updates F^t according to (14) and announces F^t to defenders;
 - 11: **end for**
 - 12: **end while**
 - 13: Each defender $i \in \mathcal{D}$ calculates x_i^* based on F^t by Theorem 3.
-

We show Algorithm 2's convergence and complexity in each iteration in the following lemma.

Lemma 5: Algorithm 2 derives the NE of the selfish filtering game. The computation and communication complexity of each iteration in Algorithm 2 is $\mathcal{O}(D)$.

Proof: See Appendix I in the supplementary material. $\square$

Lemma 5 indicates that the distributed update process converges to the NE in a finite number of iterations. We later show in experiments the fast convergence of our distributed algorithm even when the number of defenders is large.

D. Analysis of PoA

Since we use the equilibrium filtering strategy profile x^* of the selfish filtering game as the defense strategy for the edge-cloud computing network, we evaluate the effectiveness of the derived NE x^* regarding its performance on solving the total filtering cost minimization problem (5). Specifically, we analyze the price of anarchy (PoA) [35] of the selfish filtering game. In the following, we first formally define the PoA. We then prove that the PoA of the selfish filtering game always has an upper bound, indicating that the derived NE has guaranteed performance regarding the total filtering cost.

The PoA is the ratio between the total filtering cost under the NE and the minimum total filtering cost. We denote the optimal solution to the total filtering cost minimization problem (5) as x^{opt} . Then we can calculate the PoA of the selfish filtering game as

$$\text{PoA} = \frac{\sum_{i \in \mathcal{D}} C_i(x^*)}{\sum_{i \in \mathcal{D}} C_i(x^{opt})}. \quad (15)$$

It is obvious to see that $\text{PoA} \geq 1$, and a smaller PoA indicates a better performance of our derived NE x^* .

We next prove that the PoA is always upper bounded.

Theorem 5: The PoA of the selfish filtering game has an upper bound. Specifically, $1 \leq \text{PoA} < 2$ if $w_c \leq \frac{2}{\sum_{k \in \mathcal{V}} f_k}$ and $1 \leq \text{PoA} < D$ if $w_c > \frac{2}{\sum_{k \in \mathcal{V}} f_k}$.

Proof: See Appendix J in the supplementary material. $\square$

Theorem 5 shows that the PoA always has an upper bound. When the cloud latency coefficient w_c is small, the PoA is always smaller than 2. And when w_c is large, the PoA is always smaller than the number of defenders D . Note that only in exceptional cases where there is significant variation (e.g., in hundredfold) in edge servers' filtering capabilities and defenders' traffic, the PoA may approach the upper bound D .⁵ In more general cases where the heterogeneities of edge servers' filtering capabilities or defenders' traffic are smooth, the PoA is much smaller than the upper bound D . We further analyze the PoA when defenders have equal traffic.

Lemma 6: When defenders have the same amount of traffic, the PoA of the selfish filtering game is smaller than 2, i.e., $1 \leq \text{PoA} < 2$ if $\lambda_i + a_i = \lambda_j + a_j, \forall i, j \in \mathcal{D}$.

Proof: See Appendix L in the supplementary material. $\square$

Lemma 6 implies that when the traffic of all defenders is equal, the performance of our proposed distributed cooperative defense mechanism is close to the social optimal solution. Later in our simulations, we show that our proposed distributed cooperative defense mechanism can effectively reduce the total filtering cost of the edge-cloud computing network compared with existing benchmark defense mechanisms.

In this section, we have solved the defenders' cooperative defense problem in Stage II of the two-stage dynamic game, based on which we solve the DDoS attacker's attack problem in Stage I in the next section.

⁵See Appendix K in the supplementary material for an unusual example demonstrating this phenomenon.

V. OPTIMAL DDoS ATTACK STRATEGY

In Stage I of the two-stage dynamic game, the DDoS attacker strategically launches different amounts of malicious traffic a_i on different edge servers $i \in \mathcal{V}$. As shown in problem (7), the DDoS attacker aims to maximize the total filtering cost of the edge-cloud computing network, considering defenders' equilibrium filtering strategies.⁶ Recall that under an attack strategy $\mathbf{a}$, defenders fall into at most three types at the NE of the selfish filtering game as discussed in Theorem 3, and hence we can reformulate the DDoS attacker's objective function in (6) as

$$\begin{aligned} U_a(\mathbf{a}; \mathbf{x}^*) &= \sum_{0 < r_i(\mathbf{a}) < \bar{i}(\mathbf{a})} w_c(\lambda_i + a_i) + \sum_{\bar{j}(\mathbf{a}) \leq r_i(\mathbf{a}) \leq D} F^*(\mathbf{a}) \\ &+ \sum_{\bar{i}(\mathbf{a}) \leq r_i(\mathbf{a}) < \bar{j}(\mathbf{a})} \left[w_c(\lambda_i + a_i) x_{ic}^* + \sum_{k \in \mathcal{V}} x_{ik}^* F^*(\mathbf{a}) \right]. \end{aligned} \quad (16)$$

In this case, we can formulate the DDoS attacker's attack problem as follows.

$$\max_{\mathbf{a} \in \mathcal{A}} U_a(\mathbf{a}; \mathbf{x}^*). \quad (17)$$

Recall that $\mathcal{A}$ is the feasible region of the attack strategy $\mathbf{a}$ defined in Section III-B. Solving problem (17) is challenging since the objective function in Eq. (16) depends on the attack strategy $\mathbf{a}$ in a non-trivial way. Specifically, we can calculate $F^*(\mathbf{a})$ according to Eq. (11), and calculate $\bar{i}(\mathbf{a})$ and $\bar{j}(\mathbf{a})$ according to Theorem 3 which do not have closed-form solutions. Furthermore, defenders' rankings $r_i(\mathbf{a})$, $\forall i \in \mathcal{V}$ also change with the attack strategy $\mathbf{a}$, which further complicates the problem.

To solve problem (17), we first analyze a special case where all edge servers have the same amount of normal traffic. We then analyze the general case where edge servers have different amounts of normal traffic.

A. Optimal Attack Strategy When Edge Servers Have Equal Normal Traffic

We consider a special case where each edge server has the same amount of normal traffic, i.e., $\lambda_i = \lambda$, $\forall i \in \mathcal{V}$. This case happens when the network operator of the edge-cloud computing network uses load balance algorithms to distribute the same amount of requests to each edge server [36]. In this case, we first decompose problem (17) into sub-problems under fixed $(\bar{i}, \bar{j})$ pairs. We then analyze the property of each sub-problem under a fixed pair $(\bar{i}, \bar{j})$, based on which we design a low-complexity algorithm to derive the optimal attack strategy.

1) *Decomposition of Problem (17)*: The key challenge to solve problem (17) comes from the complicated relationship between the attack strategy $\mathbf{a}$ and the two critical rankings $\bar{i}(\mathbf{a})$ and $\bar{j}(\mathbf{a})$. We tackle this challenge by decomposing the

original problem (17), i.e., optimizing the objective function over the feasible region $\mathcal{A}$, into sub-problems that optimize the objective function over subsets of $\mathcal{A}$ under fixed pairs of $(\bar{i}, \bar{j})$, $\forall \bar{i} \in \{1, 2, \dots, D\}$, $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$. Specifically, we show that each fixed pair $(\bar{i}, \bar{j})$ maps to a subset $\mathcal{A}(\bar{i}, \bar{j}) \subset \mathcal{A}$ of attack strategies, and the union of all subsets $\mathcal{A}(\bar{i}, \bar{j})$, $\forall \bar{i} \in \{1, 2, \dots, D\}$, $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$ constitutes the original feasible region $\mathcal{A}$. Recall that an attack strategy $\mathbf{a}$ can map to the NE $\mathbf{x}^*$ of defenders and the corresponding pair $(\bar{i}, \bar{j})$ according to Theorem 3. Given a fixed pair $(\bar{i}, \bar{j})$ where $\bar{i} \in \{1, 2, \dots, D\}$ and $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$, we can write the subset $\mathcal{A}(\bar{i}, \bar{j})$ of the attack strategies that maps to the pair $(\bar{i}, \bar{j})$ as follows.

$$\begin{aligned} \mathcal{A}(\bar{i}, \bar{j}) &= \left\{ \mathbf{a} \mid a_i \geq 0, \forall i \in \mathcal{D}; \sum_{i \in \mathcal{D}} a_i \leq a_{\max}; \right. \\ &\frac{1}{\lambda_i + a_i} \geq \frac{w_c}{F^*(\mathbf{a})}, \forall 0 < r_i < \bar{i}; \\ &\frac{w_c}{F^*(\mathbf{a})} > \frac{1}{\lambda_i + a_i} > \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*(\mathbf{a})}, \forall \bar{i} \leq r_i < \bar{j}; \\ &\left. \frac{1}{\lambda_i + a_i} \leq \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*(\mathbf{a})}, \forall \bar{j} \leq r_i \leq D \right\}, \end{aligned} \quad (18)$$

where $F^*(\mathbf{a})$ is calculated by (11). We next prove that the union of all subsets $\mathcal{A}(\bar{i}, \bar{j})$ constitutes the original feasible region $\mathcal{A}$.

Lemma 7: The feasible region $\mathcal{A}$ of the DDoS attacker's attack strategy is the union of all subsets $\mathcal{A}(\bar{i}, \bar{j})$, i.e., $\mathcal{A} = \cup_{(\bar{i}, \bar{j})} \mathcal{A}(\bar{i}, \bar{j})$, $\forall \bar{i} \in \{1, 2, \dots, D\}$, $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$.

Proof: See Appendix M in the supplementary material. $\square$

According to Lemma 7, we can decompose the original problem (17) into sub-problems over subsets $\mathcal{A}(\bar{i}, \bar{j})$ under fixed pairs $(\bar{i}, \bar{j})$, $\forall \bar{i} \in \{1, 2, \dots, D\}$, $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$. Under a fixed pair $(\bar{i}, \bar{j})$, we can rewrite the DDoS attacker's objective function in (16) as follows.

$$\begin{aligned} U_a^{(\bar{i}, \bar{j})}(\mathbf{a}; \mathbf{x}^*) &= \sum_{0 < r_i(\mathbf{a}) < \bar{i}} w_c(\lambda_i + a_i) + \sum_{\bar{j} \leq r_i(\mathbf{a}) \leq D} F^*(\mathbf{a}) \\ &+ \sum_{\bar{i} \leq r_i(\mathbf{a}) < \bar{j}} \left[w_c(\lambda_i + a_i) x_{ic}^* + \sum_{k \in \mathcal{V}} x_{ik}^* F^*(\mathbf{a}) \right]. \end{aligned} \quad (19)$$

Then the sub-problem under the fixed pair $(\bar{i}, \bar{j})$ pair is

$$\max_{\mathbf{a} \in \mathcal{A}(\bar{i}, \bar{j})} U_a^{(\bar{i}, \bar{j})}(\mathbf{a}; \mathbf{x}^*). \quad (20)$$

To solve problem (17), we can first solve the sub-problem (20) for each fixed pair $(\bar{i}, \bar{j})$, $\forall \bar{i} \in \{1, 2, \dots, D\}$, $\bar{j} \in \{\bar{i}, \bar{i} + 1, \dots, D + 1\}$, and then choose among the optimal solutions to the sub-problems.

2) *Property of Each Sub-Problem*: For each sub-problem under a fixed pair $(\bar{i}, \bar{j})$, we first analyze its property in the special case, based on which we derive its optimal solution.

We first show that for each sub-problem, an optimal attack strategy is to launch the same amount of malicious traffic to edge servers of the same type at equilibrium (characterized in Theorem 3).

⁶Considering defenders' cooperative defense mechanism, the attack strategy that only attacking the weakest edge servers is suboptimal, since other edge servers which are not attacked can help filter the malicious traffic.

Lemma 8: For the sub-problem (20) under a fixed pair $(\bar{i}, \bar{j})$ in the special case where all edge servers have the same amount of normal traffic, the DDoS attacker can reach its optimal solution by launching the same amount of malicious traffic to defenders of the same type at equilibrium.

Proof: See Appendix N the supplementary material. $\square$

Based on Lemma 8, we denote the amount of malicious traffic launched to the three types (i.e., *cloud-only*, *mixed-filtering*, and *edge-only*) of defenders as a_c, a_m , and a_e , respectively. In other words, we have $a_i = a_c, \forall 0 < r_i < \bar{i}$, $a_i = a_m, \forall \bar{i} \leq r_i < \bar{j}$, and $a_i = a_e, \forall \bar{j} \leq r_i \leq D$. We denote $\mathbf{a}_{spe} \triangleq (a_c, a_m, a_e)$. In this case, we can rewrite the DDoS attacker's objective function in Eq. (19) under a fixed pair $(\bar{i}, \bar{j})$ as follows

$$U_{a,spe}^{(\bar{i}, \bar{j})}(\mathbf{a}_{spe}) = (\bar{i} - 1)w_c(\lambda + a_c) + (V - \bar{j} + 1)F^*(\mathbf{a}_{spe}) + (\bar{j} - \bar{i}) \left\{ w_c(\lambda + a_m) - \frac{[w_c(\lambda + a_m) - F^*(\mathbf{a}_{spe})]^2}{\lambda + a_m} \right\}.$$

We rewrite the sub-problem under a fixed pair $(\bar{i}, \bar{j})$ in problem (20) as follows

$$\begin{aligned} \max_{\mathbf{a}_{spe}} \quad & U_{a,spe}^{(\bar{i}, \bar{j})}(\mathbf{a}_{spe}) \\ \text{s.t.} \quad & (\bar{i} - 1)a_c + (\bar{j} - \bar{i})a_m + (V - \bar{j} + 1)a_e \leq a_{max}, \\ & a_c, a_e, a_m \geq 0, \\ & F^*(\mathbf{a}_{spe}) = \frac{(\bar{j} - \bar{i})w_c(\lambda + a_m) + (V - \bar{j} + 1)\frac{\lambda + a_e}{\sum_{k \in \mathcal{V}} f_k}}{\bar{j} - \bar{i} + 1}, \\ & \frac{1}{\lambda_c + a_c} \geq \frac{w_c}{F^*(\mathbf{a}_{spe})}, \\ & \frac{1}{\lambda_e + a_e} \leq \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*(\mathbf{a}_{spe})}, \text{ if } \bar{j} \neq V + 1, \\ & a_e = 0, \text{ if } \bar{j} = V + 1, \\ & \frac{w_c}{F^*(\mathbf{a}_{spe})} > \frac{1}{\lambda_m + a_m} > \frac{w_c - \frac{1}{\sum_{k \in \mathcal{V}} f_k}}{F^*(\mathbf{a}_{spe})}, \text{ if } \bar{i} \neq \bar{j}, \\ & a_m = 0, \text{ if } \bar{i} = \bar{j}. \end{aligned} \quad (21)$$

Problem (21) is a convex quadratic problem with linear constraints, and we can use Sequential Quadratic Programming (SQP) [37] to derive its optimal solution with a fast convergence speed. After solving the sub-problem for each fixed pair $(\bar{i}, \bar{j})$, we next solve the original problem (17).

3) *A Low-Complexity Algorithm to Derive the Optimal Attack Strategy:* The key idea to solve problem (17) for the special case is to derive the optimal solutions to the sub-problems under all feasible pairs $(\bar{i}, \bar{j})$, and then choose the one that can optimize the DDoS attacker's objective function.

We show the algorithm to derive the optimal attack strategy for the DDoS attacker in Algorithm 3. We first enumerate all possible $(\bar{i}, \bar{j})$ pairs (Lines 2-3). For each pair $(\bar{i}, \bar{j})$, we calculate the optimal solution $\mathbf{a}_{spe}^{(\bar{i}, \bar{j})} = (a_e^{(\bar{i}, \bar{j})}, a_c^{(\bar{i}, \bar{j})}, a_m^{(\bar{i}, \bar{j})})$ to the sub-problem (21) by SQP (Line 4). We derive the optimal solution $\mathbf{a}_{spe}^* = (a_e^*, a_c^*, a_m^*)$ to problem (17) by choosing among $\{\mathbf{a}_{spe}^{(\bar{i}, \bar{j})}, \forall i \in \{1, \dots, V\}, \bar{j} \in \{\bar{i}, \dots, V + 1\}\}$ that can maximize the attacker's objective function $U_a(\mathbf{a}_{spe}; \mathbf{x}^*)$.

We then prove the optimality of the attack strategy derived by Algorithm 3.

Algorithm 3 A Low-Complexity Algorithm to Derive the Optimal Attack Strategy for the Special Case

```

1: Set  $\mathcal{A}_{spe} = \emptyset$ ;
2: for  $\bar{i} = 1: V$  do
3:   for  $\bar{j} = \bar{i}: V + 1$  do
4:     Calculate  $\mathbf{a}_{spe}^{(\bar{i}, \bar{j})}$  by solving (21) using SQP;
5:      $\mathcal{A}_{spe} = \mathcal{A}_{spe} \cup \{\mathbf{a}_{spe}^{(\bar{i}, \bar{j})}\}$ ;
6:   end for
7: end for
8: Set  $\mathbf{a}_{spe}^* = \arg \max_{\mathbf{a}_{spe} \in \mathcal{A}_{spe}} U_a(\mathbf{a}_{spe}; \mathbf{x}^*)$ .
```

Lemma 9: Algorithm 3 derives the optimal solution to problem (17) for the special case where all edge servers have the same amount of normal traffic.

Proof: See Appendix O in the supplementary material. $\square$

Algorithm 3 can derive the optimal attack strategy in low complexity. Specifically, Algorithm 3 contains $\frac{V(V+3)}{2}$ loops. In each loop, SQP can solve the sub-problem (21) in R-quadratic convergence rate.

B. Optimal Attack Strategy When Edge Servers Have Different Normal Traffic

In this subsection, we solve the DDoS attacker's optimization problem (17) and derive the optimal attack strategy for the general case where edge servers have different amounts of normal traffic. Solving problem (17) for the general case is much more challenging than that for the special case. The reason is that even under a fixed critical ranking pair $(\bar{i}, \bar{j})$, the ranking $r_i(\mathbf{a})$ of each defender i changes with the attack strategy $\mathbf{a}$. In this case, the calculations of the subset $\mathcal{A}(\bar{i}, \bar{j})$ in Eq. (18) and the objective function $U_a^{(\bar{i}, \bar{j})}(\mathbf{a}; \mathbf{x}^*)$ in Eq. (19) under a fixed pair $(\bar{i}, \bar{j})$ have many possible outcomes. Therefore, we can not simply decompose the original problem (17) into sub-problems under each pair $(\bar{i}, \bar{j})$. In the following, we first show a property of the optimal solution, based on which we design an algorithm to solve problem (17).

The key challenge to solve problem (17) for the general case is mainly due to the change of each defender's ranking $r_i(\mathbf{a})$ with the attack strategy even under a fixed pair $(\bar{i}, \bar{j})$. We next show that there exists an optimal solution $\mathbf{a}^*$ to problem (17) under which the ranking of each defender sorted by the total traffic is the same as the ranking of each defender sorted by the normal traffic.

Lemma 10: There exists an optimal attack strategy $\mathbf{a}^*$ under which $\lambda_j + a_j^* \leq \lambda_k + a_k^*$ if $\lambda_j \leq \lambda_k, \forall j, k \in \mathcal{V}$.

Proof: See Appendix P in the supplementary material. $\square$

Lemma 10 indicates that we can optimize over the set of attack strategies under which the ranking of each defender sorted by its total traffic is the same as the ranking sorted by its normal traffic. In this case, each defender's ranking r_i does not change with the attack strategy $\mathbf{a}$ and we can decompose

problem (17) into sub-problems under each fixed pair $(\bar{i}, \bar{j})$ as follows

$$\begin{aligned} \max_{\mathbf{a} \in \mathcal{A}(\bar{i}, \bar{j})} \quad & U_a^{(\bar{i}, \bar{j})}(\mathbf{a}; \mathbf{x}^*) \\ \text{s.t.} \quad & \lambda_j + a_j \leq \lambda_k + a_k, \forall j, k \in \mathcal{V} \text{ where } \lambda_j \leq \lambda_k. \end{aligned} \quad (22)$$

Problem (22) is a non-linear problem with linear constraints. However, it's hard to analyze its convexity since its Hessian matrix is very complex. We can use SQP to calculate its KKT points as the approximate optimal solution. After deriving the approximate optimal solutions to the sub-problems under all feasible pairs $(\bar{i}, \bar{j})$, we can choose the one that maximizes the DDoS attacker's objective function.

We show the algorithm to derive the approximate optimal attack strategy for the DDoS attacker in Algorithm 4. We first enumerate all possible $(\bar{i}, \bar{j})$ pairs (Lines 2-3). For each pair $(\bar{i}, \bar{j})$, we calculate the approximate optimal solution $\mathbf{a}_{gen}^{(\bar{i}, \bar{j})} = (a_1^{(\bar{i}, \bar{j})}, a_2^{(\bar{i}, \bar{j})}, \dots, a_V^{(\bar{i}, \bar{j})})$ to the sub-problem (20) by SQP (Line 4). We derive the approximate optimal solution $\mathbf{a}_{gen}^* = (a_1^*, a_2^*, \dots, a_V^*)$ to problem (17) by choosing among $\{\mathbf{a}_{gen}^{(\bar{i}, \bar{j})}, \forall i \in \{1, \dots, V\}, \bar{j} \in \{\bar{i}, \dots, V+1\}\}$ that can maximize the attacker's objective function $U_a(\mathbf{a}_{gen}; \mathbf{x}^*)$.

Algorithm 4 A Low-Complexity Algorithm to Derive the Approximate Optimal Attack Strategy for the General Case

```

1: Set  $\mathcal{A}_{gen} = \emptyset$ ;
2: for  $\bar{i} = 1: V$  do
3:   for  $\bar{j} = \bar{i}: V+1$  do
4:     Calculate  $\mathbf{a}_{gen}^{(\bar{i}, \bar{j})}$  by solving (22) using SQP;
5:      $\mathcal{A}_{gen} = \mathcal{A}_{gen} \cup \{\mathbf{a}_{gen}^{(\bar{i}, \bar{j})}\}$ ;
6:   end for
7: end for
8: Set  $\mathbf{a}_{gen}^* = \arg \max_{\mathbf{a}_{gen} \in \mathcal{A}_{gen}} U_a(\mathbf{a}_{gen}; \mathbf{x}^*)$ .
```

We then discuss the optimality of the attack strategy derived by Algorithm 4.

Lemma 11: Algorithm 4 derives an approximate optimal solution to problem (17) by finding the optimal KKT points of all sub-problems (22).

Proof: See Appendix Q in the supplementary material. $\square$

Algorithm 4 can derive the approximate optimal attack strategy in low complexity. Specifically, Algorithm 4 contains $\frac{V(V+3)}{2}$ loops. In each loop, SQP can converge to the KKT point of sub-problem (22) in R-superlinear convergence rate.

In this section, we have solved the DDoS attacker's attack problem for both the special case and the general case. We conduct extensive simulations to evaluate the effectiveness of our derived attack strategy later in the next section.

VI. SIMULATION RESULTS

In this section, we evaluate the performance of our proposed distributed cooperative defense mechanism. We conduct simulations to answer the following questions:

Q1: How does our proposed distributed cooperative defense mechanism compare to other baseline defense methods?

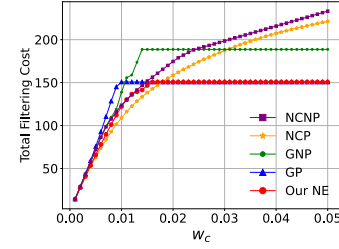


Fig. 3. Comparison of different defense methods under different values of w_c .

Q2: How do the network parameters affect the performance of our proposed defense mechanism?

Q3: How does our derived attack strategy compare to other baseline attack methods?

Q4: How does the DDoS attacker strategically behave to compromise the edge-cloud computing network?

We compare with the following baseline defense methods and baseline attack methods.

Baseline Defense Methods:

- *Non-cooperative Partial Filtering (NCP)* [12]: each defender can only partially filter its traffic on itself and on the cloud.
- *Non-cooperative Non-partial Filtering (NCNP)*: each defender can only filter its traffic either on itself or on the cloud.
- *Greedy Partial Filtering (GP)*: each defender can partially filter its traffic on any computing node to minimize its own cost in order.
- *Greedy Non-partial Filtering (GNP)*: each defender can filter its total traffic on one computing node to minimize its own cost in order.

Baseline Attack Methods:

- *Times Attack*: the amount of malicious traffic is k times of the normal traffic, and we set parameter k the same for all defenders, i.e., $k = \frac{a_{max}}{\sum_{i \in \mathcal{V}} \lambda_i}$ and $a_i = k\lambda_i, \forall i \in \mathcal{V}$.
- *Uniform Attack* [38]: the DDoS attacker launches the same amount of malicious traffic to each defender.
- *Random Attack*: the DDoS attacker launches a random amount of traffic to each defender.

In the following, we first verify the effectiveness of our proposed distributed cooperative defense mechanism (i.e., **Q1** and **Q2**). Then we evaluate the performance of our derived attack strategy (i.e., **Q3** and **Q4**).

A. Effectiveness of Proposed Defense Mechanism

In this subsection, we first compare our proposed defense mechanism with other baseline defense methods (**Q1**). Then we show the impact of network parameters on the performance of our defense mechanism (**Q2**).

1) Comparison With Baseline Defense Methods (**Q1**):

We compare our proposed distributed cooperative defense mechanism with four baseline defense methods. Specifically, we consider an edge-cloud computing network with $V = 10$ edge servers, where the amount of suspicious traffic of each edge server (i.e., $\lambda_i + a_i$) follows a uniform distribution in

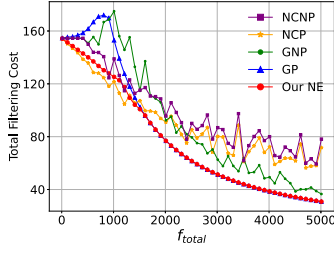


Fig. 4. Comparison of different defense methods under different values of f_{total} .

[1000, 2000] bits. We denote $f_{total} \triangleq \sum_{k \in \mathcal{V}} f_k$. Letting the filtering capability f_k of each edge server $k \in \mathcal{V}$ follow a uniform distribution in $[0, 200]$ bits/second and varying w_c in $[0, 0.05]$ seconds/bit, we get the simulation results in Fig. 3. Fixing $w_c = 0.05$ seconds/bit and varying f_{total} in $[0, 5000]$ bits/second, we get the simulation results in Fig. 4.

We show the comparison of defense mechanisms under different values of w_c in Fig. 3. We can see that when w_c is small, the NCP method can achieve the lowest total filtering cost, while our defense mechanism (i.e., the NE) achieves the second lowest total filtering cost. This is due to the limited advantage offered by edge filtering compared with the cloud scrubbing service when w_c is small. When w_c is large, our mechanism achieves the lowest total filtering cost among all defense mechanisms. That is because our defense mechanism can leverage all edge servers' filtering capabilities (compared with non-cooperative algorithms) and schedule each defender's strategies better (compared with greedy algorithms). We can also see that the total filtering cost of our mechanism remains unchanged when w_c is larger than a threshold. The reason is that in this case, filtering all the traffic on edge servers has a lower cost, and defenders will not filter their traffic on the cloud.

We show the comparison of defense mechanisms under different values of f_{total} in Fig. 4. Note that the filtering capability of each edge server is a random non-negative value, but the total filtering capability is fixed. We consider the impact of the total filtering capability because the equilibrium filtering latency F^* depends on the total filtering capability rather than the filtering capability of an edge server according to Theorem 3 and Eq. (11). Similar to Fig. 3, we can see that our defense mechanism achieves the lowest total filtering cost when the edge filtering capability is sufficient. This validates the effectiveness of our distributed cooperative defense mechanism which fully utilizes the traffic scrubbing/filtering capabilities on heterogeneous cloud and edge servers. Besides, the total filtering costs of the non-cooperative and non-partial algorithms are zigzag curves. The reason is that the filtering capability of each edge server is random under each value of f_{total} , which makes these algorithms have unstable outcomes.

2) Impact of Parameters on Defense Mechanism (Q2):

We show the defense strategy under our proposed defense mechanism, and verify the impact of network parameters on the performance of our defense mechanism.⁷

⁷The "defender index" in figures represents defenders sorted in ascending order of their total suspicious traffic (i.e., $\lambda_i + a_i$).

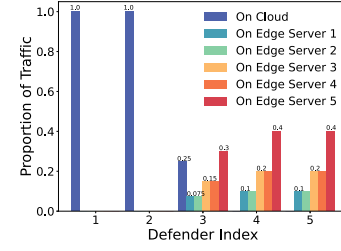


Fig. 5. Equilibrium filtering strategy.

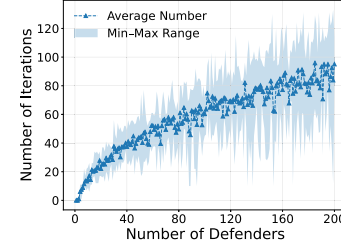
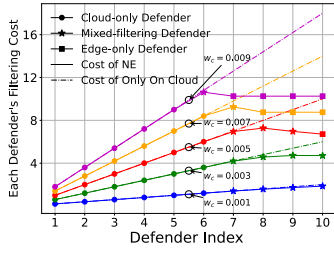
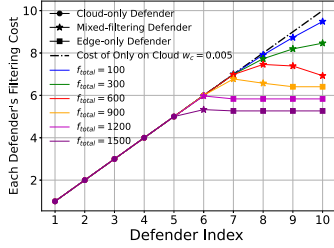


Fig. 6. Complexity of Algorithm 2 under different numbers of defenders.

We show the equilibrium filtering strategies under our proposed distributed cooperative defense mechanism in Fig. 5. We consider an edge-cloud computing network with 5 edge servers whose total suspicious traffic (in bits) are 100, 500, 1000, 1500 and 2000, respectively, and the filtering capability (in bits/second) of the edge servers are 50, 50, 100, 100 and 200, respectively. We set $w_c = 0.01$ s/bit. In this case, we can calculate by Algorithm 1 that the equilibrium filtering latency is $F^* = 8.5$. Fig. 5 shows that at equilibrium, defender 1 and defender 2 are *Type I (cloud-only)* defenders, because the marginal filtering cost on the cloud is lower than the marginal filtering cost on edge servers. Defender 3 is a *Type II (mixed-filtering)* defender since the marginal filtering cost on the cloud is equal to that on edge servers. Defender 4 and defender 5 are *Type III (edge-only)* defenders, because filtering on the cloud for them has a greater marginal filtering cost than filtering only on edge servers. The simulation results are consistent with our theoretical analysis in Theorem 3.

To show the scalability of our proposed distributed cooperative defense mechanism, we plot the number of iterations of Algorithm 2 under different numbers of defenders D in Fig. 6. For a fixed number of defenders, we conduct 10 simulations (where in each simulation, each defender's amount of suspicious traffic follows a uniform distribution in $[1000, 2000]$). We set the filtering capability of each defender as 200 bits/s and $w_c = 0.0001$ s/bit. We set the tolerance of the algorithm as $\epsilon = 10^{-6}$. We can see from Fig. 6 that while the number of iterations exhibits some variation, the average number of iterations is always smaller than D , indicating that our defense mechanism can respond to the DDoS attack quickly. This is consistent with our complexity of the Algorithm 2. The fluctuations of the curve are due to the shifting distance between the equilibrium filtering latency F^* and the initial point, which is influenced by changes in the critical pair $(\bar{i}, \bar{j})$.

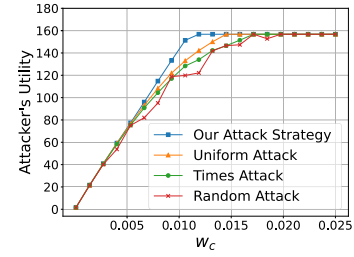
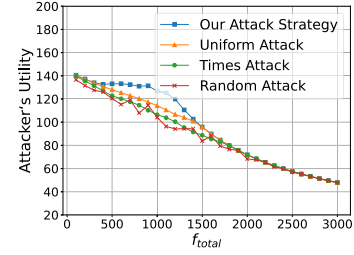
Fig. 7. Defenders' filtering costs under different values of w_c .Fig. 8. Defenders' filtering costs under different values of f_{total} .

We then verify the impact of the network operators, i.e., w_c and f_{total} , on the equilibrium defense strategies. Specifically, we consider an edge-cloud computing network with $V = 10$ edge servers whose amount of suspicious traffic (i.e., $\lambda_i + a_i$) is 200, 400, $\dots$, 2000 bits, respectively. We fix $f_{total} = 700$ bits/second and vary w_c , and get Fig. 7. We fix $w_c = 0.005$ seconds/bit and vary f_{total} , and get Fig. 8.⁸

Fig. 7 shows defenders' filtering costs under different values of w_c . We can see that as w_c increases, defenders' filtering costs of all defenders increase. Furthermore, more defenders filter their traffic on edge servers at equilibrium, and fewer defenders filter their traffic on the cloud due to the large cloud filtering costs. For example, when $w_c = 0.001$ s/bit, defenders indexed from 7 to 10 are all *mixed-filtering* defenders. When $w_c = 0.009$ s/bit, defenders indexed from 7 to 10 all change to be *edge-only* defenders. An interesting phenomenon is that under a fixed w_c , the filtering cost of *mixed-filtering* defender is larger than the filtering cost of *edge-only* defender. The reason is that at equilibrium, the defenders with more traffic filter more traffic on edge servers and defenders with less traffic are forced to filter their traffic on the cloud due to congestion on edge servers. For *mixed-filtering* defenders, their filtering costs are larger since the filtering cost on the cloud is large.

Fig. 8 shows defenders' filtering costs under different values of f_{total} . We can see that as the total filtering capability f_{total} increases, more defenders filter their traffic on edge servers at equilibrium, and defenders' filtering costs decrease. The reason is that when edge servers have larger filtering capabilities, the congestion effect due to resource sharing among defenders is effectively alleviated, and the traffic filtering latency on edge servers is smaller.

⁸To show the type of each defender at equilibrium, we use circle, star and square to represent defenders of Type I (cloud-only), Type II (mixed-filtering), and Type III (edge-only). The solid line represents each defender's filtering cost under our defense mechanism, and the dotted line represents each defender's filtering cost if it only filters traffic on the cloud.

Fig. 9. Comparison of different attack methods under different values of w_c .Fig. 10. Comparison of different attack methods under different values of f_{total} .

Both Fig. 7 and Fig. 8 demonstrate the superiority of our defense mechanism compared with solely utilizing cloud scrubbing. Furthermore, our defense mechanism proves advantageous for heavy-traffic defenders.

B. Performance of Derived Attack Strategy

In this subsection, we first compare our derived attack strategy with other baseline attack methods. Then we show the attacker's strategic attack strategies in different cases.

1) *Comparison With Baseline Attack Methods (Q3)*: To justify that our attack strategy can model the attacker's strategic behaviors better, we compare our attack strategy derived by Algorithm 4 with three baseline attack methods. We consider an edge-cloud computing network with $V = 10$ edge servers whose amount of normal traffic (i.e., λ_i) follows a uniform distribution in [500, 1500] bits. The total amount of malicious traffic that the DDoS attacker can launch to the edge-cloud computing network is $a_{max} = 5000$ bits. Fig. 9 shows the comparison of different attack methods under different values of w_c when $f_{total} = 1000$ bits/s. Fig. 10 shows the comparison of different attack methods under different values f_{total} when $w_c = 0.01$ s/bit. Both figures show that under our attack strategy, the DDoS attacker achieves the largest utility, where the utility increase can be up to 27.7%. The reason is that our attack strategy optimization problem allows the DDoS attacker to strategically launch different amounts of malicious traffic to different edge servers, and further considers the defense strategy of the edge-cloud computing network. Note that when w_c or f_{total} increases to a certain value, all defenders choose to be *edge-only* defenders at equilibrium and filter all their traffic proportionally on edge servers. In this case, different attack methods achieve the same performance due to the same equilibrium filtering latency F^* .

2) *Attacker's Strategic Attack Behaviors (Q4)*: We show the DDoS attacker's optimal attack strategy for both the special case and the general case.

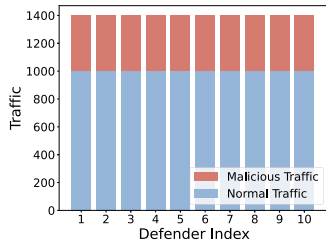


Fig. 11. Optimal attack strategy in special case.

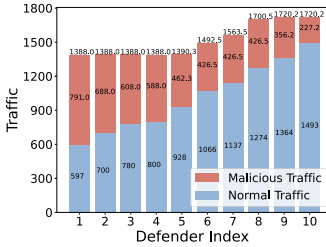


Fig. 12. Approximate optimal attack strategy in general case.

Fig. 11 shows the DDoS attacker's optimal attack strategy for the special case. We consider an edge-cloud computing network with 10 edge servers that have equal normal traffic of 1000 bits. We assume that the total amount of malicious traffic (in bits) that the DDoS attacker can launch to the edge-cloud computing network is $a_{max} = 4000$ bits. We set $w_c = 0.01$ s/bit and $f_{total} = 200$ bits/s. In this case, the DDoS attacker's optimal attack strategy derived by Algorithm 3 is $a_c^* = 0$, $a_m^* = 400$, $a_e^* = 0$. In other words, all defenders choose to be *mixed-filtering* defenders at equilibrium, and the DDoS attacker evenly distributes its malicious traffic to each defender. Based on Lemma 6, the PoA is less than 2 when defenders' total suspicious traffic is equal. This indicates that our defense mechanism can achieve effective defense performance.

Fig. 12 shows the approximate optimal attack strategy derived by Algorithm 4 for the general case. We consider an edge-cloud computing network with 10 edge servers whose normal traffic follows a uniform distribution in $[500, 1500]$. We set $a_{max} = 5000$ bits, $w_c = 0.001$ s/bit, and $f_{total} = 5000$ bits/s. We can see from the Fig. 12 that defenders indexed from 1 to 4 have the same amount of total traffic under attack, and defenders indexed 9 and 10 also have identical total traffic. In comparison with the original normal traffic, the total traffic of each defender under attack becomes closer, as evidenced by a decrease in standard deviation from 285.97 to 141.96. Experiments under different parameter settings show that the standard deviation of the total traffic is lower than that of the normal traffic. As discussed in Section IV-D, the PoA is small when edge servers' total traffic does not exhibit significant variations. Therefore, our defense mechanism can still effectively mitigate the strategic DDoS attacks in general cases.

VII. CONCLUSION

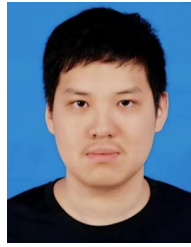
In this paper, we design a distributed cooperative defense mechanism for edge-cloud computing networks considering

the DDoS attacker's strategic attack behavior. We model the interactions between the DDoS attacker and defenders as a two-stage dynamic game. We solve the total filtering cost minimization problem in Stage II by modeling defenders' traffic filtering behaviors as a selfish filtering game, for which we prove the uniqueness of the Nash equilibrium. By utilizing the property that the equilibrium filtering latency on each edge server is the same, we design a distributed algorithm for each defender to derive its equilibrium filtering strategy without knowing the global information. We further prove that the PoA of the derived NE has an upper bound, indicating guaranteed social efficiency. For the DDoS attacker's optimal attack problem in Stage I, we design low-complexity algorithms to derive the optimal attack strategies. Simulation results show that the DDoS attacker tends to attack all edge servers to reduce their cooperative defense capability, and our defense mechanism can achieve a better defense performance compared with existing benchmark mechanisms. For future work, it would be interesting to study defense mechanisms under more general attacker models (e.g., boundedly rational, heuristic, or adaptive attackers) and dynamic system parameters.

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